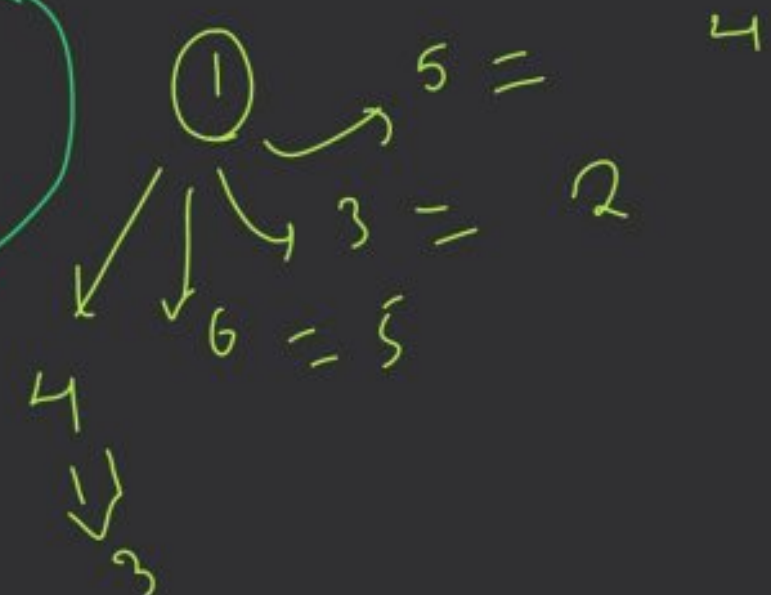


Leetcode 121:- Best Time to Buy & Sell Stock I:-

$[7, 1, 5, 3, 6, 4] = \text{prices}[i]$

$[4, 3, 2, 1] \times$



int minm = INT_MAX;
int profit = 0;

for (i = 0 → n) {
if (prices[i] < minm) {
minm = prices[i];
}
profit = max(profit, prices[i] - minm);
}

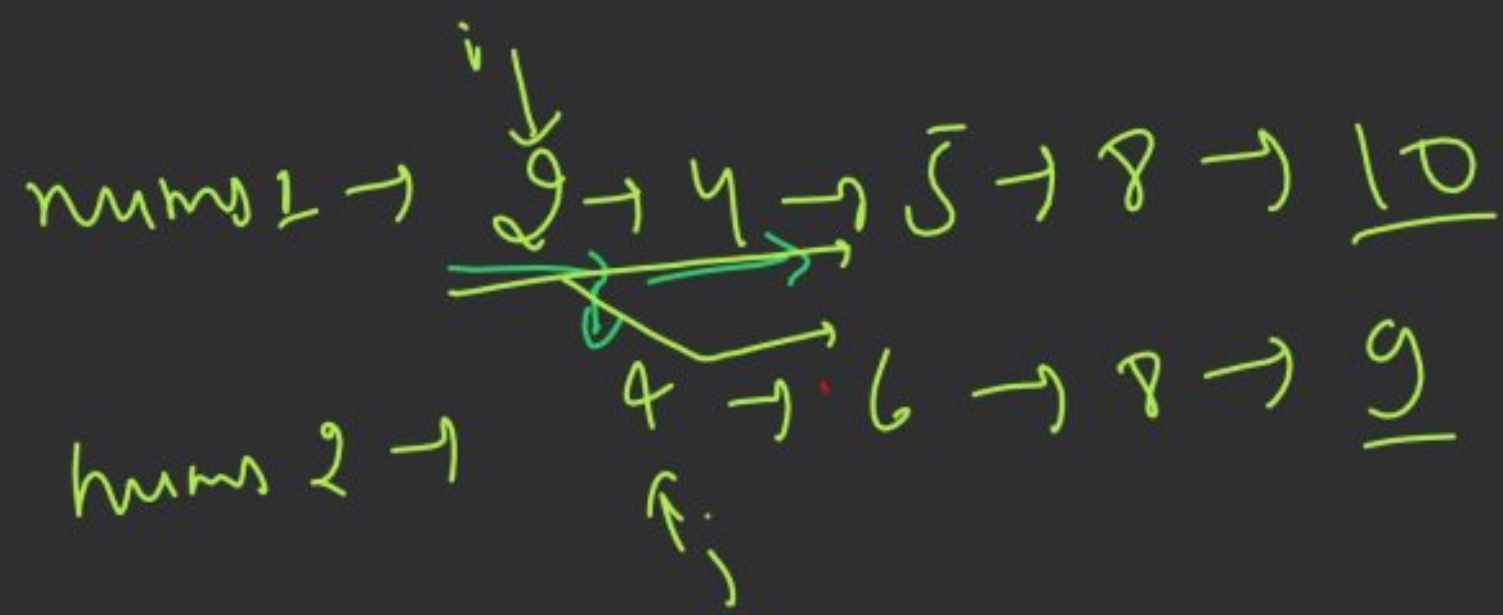
0, 10 - 2
8

(8, 1, 0, 1)

$[6, 2, 10, 1, 8, 7, 100]$

minm = INT_MAX, 6, 2, 1
profit = 0, 0, 0, 1, 99

Ques.



S-I :- (Before first common = 4)

nums1 : 2 sum1 = 2
nums2 : sum2 = 0

At 4, $\max(2, 0) + 4 = 2 + 4 = 6$

S-II :- (Before 8)

nums1 : 5

nums2 : 6

At 8, $\max(5, 6) + 8 = 14$

S-III :- after 8 :-

nums1 : 10

nums2 : 9

$\max(10, 9) = 10$

(9, 10)

num 1 = [1, 3, 5, (7, 9)]

num 2 = [1, 3, 5]
↑ ↑ ↑
j i i

$$\max(0, 0) + 1 = 0 + 1 = 1$$

$$1 + 3 = 4$$

$$4 + 5 = 9$$

$$\max(120, 120) + 10$$

↓
130 - 110 = 20 → result

[50, 80, 10, 9, 3, 10, 100]

[20, (100), 10, 200, 10, 10, 1]

80 < 100 ✓ Sum1 = 50 130
50 > 20 ✓ Sum2 = 20 (120)

(100 > 10)

> < =

✓ fail

(Goal is necessary)

[1, 2, 10, 100] (2 < 50)

[1, 50, 60, 100]

i++
j++

result = 4

sum1 = 2

sum2 = 50/100

[1, 3, 5, 7, 9]

[1, 3, 5]

→ 1, 2, 3, 10
→ 5, 9, 10

sum1 = 786

sum2 = (23) + 12 = 35

i=0, j=0, sum1=0, sum2=0, result=0;

while (i < num1.length && j < num2.length) {
num1[i] == num2[j]

if (num1[i] < num2[j]) {

sum1 += num1[i];

i++;

else if (num1[i] > num2[j]) {

sum2 += num2[j];

j++;

}

else {

result = result + max

(sum1, sum2)
+ num1[i],

sum1 = 0;

sum2 = 0;

i++, j++;

}

~~[1, 2, 70, 100]~~

70 < 60

(2 < 50)

[⁰1, ¹50, ²60, ³100, ⁴1, 2, 3, 4, 5, 6, 7, 8, 9]

~~i~~ ~~j~~ Sum1 = ~~0~~ ~~2~~ 72

Sum2 = ~~0~~ ~~50~~ 110

res = ~~0~~ ~~2~~ 110

max(Sum1, Sum2) + nums2(i)

res = 110 + 100 = 210

i > j
= = ✓

nums2(j) < nums1(i)

or

nums1(i) > nums2(j)

70 < 60

20 < 10

(Sum2)

while(i < nums1.length)
Sum1 += nums1[i];
while(j < nums2.length)
Sum2 += nums2[j];

Majority Element :-

mony ↓

Voting Element

count, el = nums[0];
= 1;

for (i = 1 → n) {

if (nums[i] == el)

count++;

else

[5, 2, 3, 5, 5, 5, 10, 5, 5, 10]

el = 5 2 3 10 10

count = 1 0 1 2 2 2 2 (4)

[9, 9, 9, 5, 5, 5, 2, 5, 2, 2]

int count = 0, el;

if (count == 0) el = nums[i];

if (nums[i] == el) count++;

else count--;

(2, 5, 5)

[1, 1, 1, 5, 5, 5, 5, 5, 5]

count = 0 1 2 3 2 1 0 3

el = 0 1 (5)

count = 0 1 2 3 2 1 0

el = 0 2 2